1 Introduction

1.1 Background

1.1.1.1 In order to address the demand for land for housing, the Policy Addresses (i.e. from Year 2012 to Year 2015) announced the review of agricultural land in the North District and Yuen Long, which is mainly deserted or being used for industrial purposes and/or temporary storage purposes, as one of the land supply measures. Furthermore, the recent Policy Addresses in Year 2016 and Year 2017 also mentioned Yuen Long South (YLS) development as a medium and long-term land supply measure.

1.1.1.2 To optimise the use of degraded brownfield sites so as to release their development potential in meeting the territory's medium to long-term housing needs, the Planning and Engineering Study for the Housing Sites in YLS – Investigation (the Study) commenced in November 2012 to examine the future land use, optimise the development potential of the degraded brownfield, improve existing degraded environment and ascertain the feasibility for public and private housing developments and other uses within the YLS Potential Development Area (PDA)[1] with supporting infrastructure and community facilities (The Project).

1.1.1.3 The YLS PDA is located to the south of Yuen Long New Town, and in the proximity of Tin Shui Wai New Town and the planned Hung Shui Kiu (HSK) New Development Area (NDA). The area is connected with the urban areas by strategic road links, including Route 3 and Yuen Long Highway (YLH).

1.1.1.4 The PDA, which has an area of 223.5 hectares (ha), is surrounded by YLH at the north and Tai Lam Country Park at the south. The PDA includes the Tong Yan San Tsuen (TYST) area, a piece of land zoned "Undetermined" ("U") on TYST Outline Zoning Plan (OZP) as well as two small pieces of land in the Tai Tong area which are zoned "Other Specified Uses" annotated "Rural Use" ("OU(RU)") on Tai Tong OZP.

1.1.1.5 The YLS PDA is currently rural in character with a mixture of land uses. The predominant uses are brownfield operations including open storage yards, warehouses, industrial workshops, etc. These brownfield operations are intermingled with rural settlements and residential developments, agricultural land, livestock farms and vacant land. Proliferation of such brownfield sites has resulted in degradation of the rural environment due to noise nuisances to surrounding residents, and discharge of industrial waste water, i.e. from daily cleaning at the working area, waste water from workshops after heavy rainfall, etc. Furthermore, any of these industrial premises which may not be properly connected to sewer system could result in discharge of untreated sewage into nearby water bodies and hence deteriorate water quality.

1.1.1.6 There is a need to optimise the use of some degraded brownfield sites for beneficial purposes so as to create development potential to meet the territory's medium to long-term housing needs, and improve the existing environment. The Study is aimed to examine the future land use, optimise the development potential, and ascertain the

[1] The "Potential Development Area" in this EIA Report is the same as the "Development Area" referred in the Recommended Outline Development Plan.

feasibility for public and private housing developments and other uses within YLS PDA.

1.2 The Study

1.2.1.1 Planning Department (PlanD) and Civil Engineering and Development Department (CEDD) of the HKSAR commissioned Ove Arup and Partners Hong Kong Limited (Arup) in 2012 to undertake the Planning and Engineering Study for Housing Sites in YLS – Investigation (the Study). The Study will examine the future land use, optimise the development potential, and ascertain the feasibility for public and private housing developments in the YLS PDA. Specifically, the Study will:

- examine and identify sites within the YLS PDA for public and private housing developments, supporting Government, Institution or Community (G/IC) facilities, open space and/or amenities and other uses, and review the boundaries of the PDA;
- recommend appropriate development parameters for the development sites;
- ascertain the feasibility and acceptability of the Study proposals in terms of traffic and other infrastructure capacities, urban design, environment, air ventilation, etc.;
- identify and propose engineering infrastructure works that are needed to support the Study proposals; and
- undertake a three-stage Community Engagement (CE) Programme to solicit public views on the Study proposals within the PDA as input to the Study.

1.2.1.2 The findings and recommendations of the Study will serve as a reference for the revision of the TYST OZP, Tai Tong OZP and the prevailing OZPs that will be affected by the development proposals to guide the YLS developments and engineering infrastructure upgrading works identified.

1.3 Description of Project

1.3.1.1 The YLS PDA covers approximately 223.5 hectares (ha) and is located to the south of Yuen Long New Town and bounded by YLH, Kung Um Road and Tai Lam Country Park. The PDA is divided into three areas: the area to the western side that includes TYST (Area 1), the northern part of the land that lies on both sides of Kung Um Road and Kiu Hing Road (Area 2), and the southern part of the land that lies on both sides of Kung Um Road and Pak Sha Shan Road (Area 3). Other than development within the PDA, there are supporting infrastructure works outside the PDA boundaries, which include the following:

- New slip roads connecting the PDA to Pok Oi Interchange and a new PTI next to Yoho Midtown ;
- New PTI next to Yoho Midtown;
- New TYST Reclaimed Water Service Reservoir and its associated works;
- Partial decking of Yuen Long Nullah along Kung Um Road and Kiu Hing Road;
- Modification works on the trunk roads connecting Tin Shui Wai West (TSWW) Interchange;

- Modification works on the primary distributor roads connecting TYST Interchange;
- Other new local roads connecting PDA;
- Rising Main for water supplies and sewage; and
- New sewer from YLS STW to existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall.

1.3.1.2 The location of Project is shown in **Figure 1.1**.

1.4 EIA Study Brief

1.4.1.1 In accordance with the requirements of Section 5(1)(a) of the Technical Memorandum – Environmental Impact Assessment Ordinance (EIAO-TM), an application (No. ESB-246/2012) for an Environmental Impact Assessment (EIA) Study Brief was submitted to Environmental Protection Department (EPD) on 7 May 2012 with a Project Profile (No. PP-464/2012). Pursuant to Section 5(7)(a) of the EIA Ordinance, the Director of Environmental Protection issued to the Project Proponent, namely CEDD, an EIA Study Brief (ref: EIA Study Brief No: ESB-246/2012 dated 18 June 2012) to carry out an EIA Study.

1.5 Designated Projects

1.5.1.1 The Study is a Designated Project (DP) under Item 1 Schedule 3 of EIAO - Engineering feasibility study of urban development projects with a study area covering more than 20 ha or involving a total population of more than 100,000. To implement the Project, there are various key items among most of which are classified as DPs under Schedule 2 of the EIA Ordinance. A list of Schedule 2 DPs is summarised in **Table 1.1** below and these DPs are further described in the sequent sections. The locations of these DPs under the Project are shown in **Figure 1.2**.

Table 1.1 Summary of schedule 2 DPs

DP Reference No.	Ref. Category No.	Designated Projects	Ref. in Recommended Outline Development Plan (RODP)	Works responsible by Project Proponent	EP application under the Study
DP1	Part I, A.1	A road which is an expressway, trunk road, primary distributor road or district distributor road including new roads, and major extensions or improvements to existing road	Construction of Trunk Roads (TSWW Interchange)	• Site formation ✓ • Construction ✓ • Operation X	Y
DP2	Part I, A.1	A road which is an expressway, trunk road, primary distributor road or district distributor road including new roads, and major extensions or improvements to existing road	Construction of new Primary Distributor Roads (TYST Interchange)	• Site formation ✓ • Construction ✓ • Operation X	Y
DP3	Part I, A.1	A road which is an expressway, trunk road, primary distributor road or district distributor road including new roads, and major extensions or improvements to existing road	Construction of two new Distributor Roads (Road D1 to Road D2)	• Site formation ✓ • Construction ✓ • Operation X	Y
DP4 [1] (By others)	Part I, A.3	A tramway and its associated stations	Environmentally Friendly Transport Services (EFTS)	• Site formation ✓ • Construction X • Operation X	N
DP5	Part I, A.8	A road or railway bridge more than 100m in length between abutments	Construction of slip roads at the TYST Interchange	• Site formation ✓ • Construction ✓ • Operation X	Y
DP6	Part I, A.9	A road fully enclosed by decking above and by structure on the sides for more than 100m	Construction of partly depressed road/ underpass located at TSWW Interchange and full enclosures at TYST Interchange	• Site formation ✓ • Construction ✓ • Operation X	Y
DP7 [1] (By others)	Part I, F.2	Sewage treatment works with an installed capacity of more than 5,000m^3 per day and located at less than 200m from existing / planned receivers	Construction of new YLS STW (Site OU3.3)	• Site formation ✓ • Construction X • Operation X	N

DP Reference No.	Ref. Category No.	Designated Projects	Ref. in Recommended Outline Development Plan (RODP)	Works responsible by Project Proponent	EP application under the Study
DP8	Part I, F.3(b)	A sewage pumping station with an installed capacity of more than 2,000m^3 per day and located at less than 150m from existing / planned receivers	Construction of two new SPSs (Sites OU1.12 and OU2.2)	• Site formation ✓ • Construction ✓ • Operation X	Y
DP9 [1] (By others)	Part I, F.4	An activity for the reuse of treated sewage effluent from a treatment plant	Construction of Reclaimed Water Service Reservoir for reuse of reclaimed water	• Site formation ✓ • Construction X • Operation X	N
DP10	Part I, I.1(b)(vii)	A drainage channel or river training and diversion works less than 300m from the nearest boundary of an existing conservation area	Yuen Long Nullah revitalisation/decking along Kung Um Road and Kiu Hing Road	• Site formation ✓ • Construction ✓ • Operation X	Y

Notes:

[1] The project element is a separate DP to be constructed and operated by the future relevant project proponents. The future relevant project proponents shall conduct separate studies, including EIA for approval under the EIA Ordinance if necessary, to determine its feasibility and implementation programme. However, the site formation work for this DP will be covered in this EIA

DP 1, 2 and 3 Part I, A.1 – Construction of trunk roads, primary distributor roads and district distributor roads

1.5.1.2 The major objective of the proposed road network is to improve connectivity with Yuen Long New Town, the proposed HSK NDA, and provides a direct access and egress to YLH. Long Hon Road is proposed to be upgraded and extended to Kung Um Road to form a key corridor of the PDA.

1.5.1.3 The road system will provide direct and convenient access of freight traffic between the Employment Belt in Area 1 and YLH without travelling through the residential neighbourhood. The TSWW Interchange (trunk roads) is improved to increase traffic movements and cater for direct access between the Employment Belt and YLH through the westbound carriageway.

1.5.1.4 The improved TYST Interchange (primary distributor roads) will be modified to cater for all traffic movements to and from YLH, Long Tin Road and Long Hon Road.

1.5.1.5 In addition, district distributor Road D1 will connect through TYST Interchange and Area 1 to Area 2 and 3 of the PDA, and district distributor Road D2 will connect Kung Um Road, Lam Tai West/East Road to Road D1.

1.5.1.6 In addition, a pair of slip roads (local distributor roads) running on both sides and parallel with YLH from Kung Um Road eastward are proposed to provide connectivity to YLH via Shap Pat Heung Interchange. These roads are local distributor roads which are not DPs.

1.5.1.7 For the northern section of Kung Um Road and Kiu Hing Road (local distributor roads) along Yuen Long Nullah, road widening will be undertaken through partial decking of Yuen Long Nullah. For Kung Um Road, the existing two lanes will be kept and changed to one-way direction, with minor road improvement works including road widening and footpath provision. For Kiu Hing Road, the existing one lane road will be upgraded as standard one-way 2-lane single carriageway. This proposal allows provision of adequate road space for vehicles and pedestrians. These roads are local distributor roads which are not DPs.

1.5.1.8 Since the proposed works involve construction and major improvements of trunk roads, primary distributor roads and district distributor roads, they will be DPs under Item A.1 of Schedule 2, Part I of EIA Ordinance.

DP 4 - Part I, A.3 - Construction of Environmentally Friendly Transport Services (EFTS) – subject to further review

1.5.1.9 The RODP has reserved a corridor for the extension of the EFTS proposed under the HSK NDA project with possible connection to West Rail Tin Shui Wai Station. The reserve navigates through the storage/workshops, residential, commercial and other land reserves within the PDA. As mentioned in the approved HSK EIA report, a railway transport system has been assumed for the purpose of the EIA. However, the EFTS system is a separate DP to be constructed and operated by the future operator who would need to prepare and submit a separate EIA Study for approval under the EIA Ordinance.

DP 5 - Part I, A.8 – Construction of a road bridge more than 100m in length between abutments

1.5.1.10 As part of the road improvement works, an elevated viaduct section of more than 100m in length are proposed at TYST Interchange in the RODP.

1.5.1.11 Since the proposed works involve construction of a road bridge more than 100m in length between abutments in the RODP, it will be a DP under Item A.8 of Schedule 2, Part I of EIA Ordinance.

DP 6 - Part I, A.9 – Construction of a road fully enclosed by decking above and by structure on the sides for more than 100m in length

1.5.1.12 As part of the TSWW Interchange improvement works, a depressed road/ underpass of more than 100m in length beneath the at-grade roundabout at the TSWW Interchange is proposed in the RODP. In addition, full enclosures at TYST Interchange are also required in order to comply the noise criteria.

1.5.1.13 Since the proposed works involve construction of a fully enclosed road of more than 100m in length in the RODP, it will be a DP under Item A.9 of Schedule 2, Part I of EIA Ordinance.

DP 7 - Part I, F.2 – Construction of a Sewage Treatment Works – subject to further review

1.5.1.14 With other major developments including HSK NDA, Kam Tin South and housing sites at Wang Chau, there is neither spare sewage treatment capacity at the nearby San Wai Sewage Treatment Work (SWSTW) nor Yuen Long Sewage Treatment Works (YLSTW) for the YLS development, despite planned upgrading of these sewage treatment works (STWs). There is also little scope to further expand these two STWs after discussions with the relevant authorities.

1.5.1.15 An "OU(STW)" zone at the southern end of Kung Um Road is reserved for the construction of an aboveground STW (YLS STW) for treatment and disposal of sewage arising from the PDA. Based on the sewage estimation according to the RODP, the treatment capacity of YLS STW would have an Average Dry Weather Flow (ADWF) of 24,000m^3/day.

1.5.1.16 Since the proposed works involve construction of sewage treatment works with installed capacity more than 5,000m^3 per day, and they are located close to the planned receivers in the RODP, it will be a DP under Item F.3(b) of Schedule 2, Part I of EIA Ordinance. However, the STW is a separate DP to be constructed and operated by the future operator who would need to prepare and submit a separate EIA for approval under the EIA Ordinance.

DP 8 - Part I, F.3(b) – Construction of two sewage pumping stations

1.5.1.17 Two "OU(RCP&SPS)" zones are intended for the sewage pumping stations (SPSs) and refuse collection points (RCPs).

1.5.1.18 The SPS site located to the east of TYST Interchange in Area 1 is reserved for an intermediate SPS to pump the sewage collected from the Area 1 for further discharge by gravity sewerage to the YLS STW for treatment, as well as an RCP. The ADWF of the SPS would be approximately 4,700m^3/day.

1.5.1.19 The other SPSs site located to the west of the northern end of Kung Um Road in Area 2 is reserved to retain the existing Lung Tin SPS, construct a SPS to pump the sewage collected from the Areas 2 and 3 by underground gravity sewers to the YLS STW for treatment, as well as an RCP. The ADWF of the SPS would be approximately 23,280m^3/day at the northern end of Kung Um Road.

1.5.1.20 Since the proposed works involve construction of SPS with installed capacity more than 2,000m^3 per day, and they are located close to the planned receivers in the RODP, it will be a DP under Item F.3(b) of Schedule 2, Part I of EIA Ordinance.

DP 9 - Part I, F.4 - An activity for the reuse of reclaimed water from a treatment plant – subject to further review

1.5.1.21 An area at the southern end of TYST is reserved for the construction of a Reclaimed Water Service Reservoir for reuse of reclaimed water generated from the YLS STW.

1.5.1.22 Since the proposed works involve reuse of reclaimed water from a treatment plant, it will be a DP under Item F.4 of Schedule 2, Part I of EIA Ordinance. However, the reuse activity is a separate DP to be constructed and operated by the future operator who would need to prepare and submit a separate EIA for approval under the EIA Ordinance.

DP 10 - Part I, I.1(b)(vii) - Yuen Long Nullah Revitalisation

1.5.1.23 Yuen Long Nullah will be partially decked to provide space for road widening and provision of landscaped footpaths at Kung Um Road and Kiu Hing Road. The nullah will also be reviatlised and integrated in the urban design and landscape framework.

1.5.1.24 Since the proposed works involve a drainage channel or river training less than 300m from the nearest boundary of an existing Conservation Area under TYST OZP Plan No. S/YL-TYST/10, it will be a DP under Item I.1(b)(vii) of Schedule 2, Part I of EIA Ordinance.

1.5.1.25 This EIA study will therefore cover all the Schedule 2 DPs mentioned except DP4, DP7 and DP9 in **Table 1.1**.

1.6 Non Designated Projects

1.6.1.1 Apart from the Schedule 2 DPs as listed in **Table 1.1**, there are also a number of non-DP elements in the RODP including but not limited to the following sub-elements summarised in **Table 1.2**. Details of these non DP elements are presented in **Chapter 3** and their locations and lot reference are shown in **Figure 3.1**.

Table 1.2 Summary of non-DPs

Non DPs	Sub-elements
Residential Uses	• Special Residential – Public Rental Housing (with commercial) ("RSc") • Residential - Zone 1 (Subsidised Sale Flats) (with commercial) ("R1(SSF)c"); • Residential - Zone 2 (Subsidised Sale Flats) (with commercial) ("R2(SSF)c") • Residential - Zone 1 (with commercial) ("R1c") • Residential - Zone 2 (with commercial) ("R2c") • Residential - Zone 2 ("R2") • Residential - Zone 3 ("R3") • Residential - Zone 4 ("R4") • Residential - Zone 5 ("R5") • Residential - Zone 5 (Existing Development Area ("R5(EDA)") • Residential - Zone 6 ("R6") • Other Specified Uses (Mixed Use) ("OU(MU)")
Storage and Workshop Uses	• Other Specified Uses (Open Storage) ("OU(OS)") • Other Specified Uses (Storage Use) ("OU(SU)") • Other Specified Uses (Storage + Workshop) ("OU(S+W)")
Government, Institution or Community	• Government ("G") • Institution or Community ("IC")
Education	• Education ("E")
Public Utilities ("OU")	• Other Specified Uses (Sewage Treatment Works) ("OU(STW)") • Other Specified Uses (Refuse Collection Point and Sewage Pumping Station) ("OU(RCP&SPS)") • Other Specified Uses (Electricity Substation) ("OU(ESS)") • Other Specified Uses (Service Reservoir) ("OU(Service Reservoir)") • Other Specified Uses (Hillside River Corridor with Scenic Cycle Track) ("OU(Hillside River Corridor with Scenic Cycle Track)") (1)
Road and River Channels	• Local Roads (L1 – L27) • Lam Tai East Road • Lam Tai West Road • Kung Um Road • Kiu Hing Road • Tam Kwai Tsuen Road • Lam Yu Road • Tai Kei Leng Road • Long Ho Road • Yuen Long West Nullah Revitalisation • Tin Tsuen Channel Revitalisation
Amenity	• Amenity ("A")
Open Space	• District Open Space ("DO") • District Open Space (1) ("DO (1)") • Local Open Space ("LO") • Local Open Space (1) ("LO (1)")
Agriculture	• Agriculture ("AGR")
Green Belt	• Green Belt ("GB") • Green Belt (1) ("GB(1)")
Village Type Development	• Village Type Development ("V")

Note:
(1) "OU (Hillside River Corridor) with Scenic Cycle Track" also includes reedbed/retention pond.

1.7 Green Initiatives, Project Benefits and Environmental Initiatives

Green Initiatives

1.7.1.1 The vision of the Project is to create a sustainable, green and liveable community provided with supporting infrastructure to cater for future development needs.

1.7.1.2 With the increasing awareness on sustainable development, green building design, energy-efficient features and renewable energy technologies within the PDA have also been duly considered. The Project has adopted a sustainable strategy in the guiding principles, in respect of town planning, urban design, transportation and blue-green infrastructure for a creation of a sustainable local community. Within the PDA, green mobility is promoted through the reservation of space for EFTS, and comprehensive pedestrian walkways and cycle track networks.

1.7.1.3 In addition to green transport management, a comprehensive water management system including sewerage, drainage and water resources infrastructure has been recommended. To promote sustainable use of water, opportunities for reuse of TSE as reclaimed water for non-potable uses such as toilet flushing within the PDA and adjacent developments have been considered. Blue-green infrastructure, such as the retention tank, reedbed/ retention pond, retention lake and Hillside River Corridor to collect runoffs from terrains, will be used for temporary storage of stormwater.

1.7.1.4 As part of the urban design initiative, sections of the existing Yuen Long (West) Nullah, Tin Tsuen Channel, and Yuen Long Nullah are proposed to be revitalised. After the revitalisation, these nullahs would provide an aesthetic environment for the local community. The nullahs will provide a rural, natural sensation in the urban community and act as key north-to-south connectivity corridors together with provisions of pedestrian walkway and cycle track networks.

1.7.1.5 Subject to further review, the above green initiatives will be implemented for future developments.

1.7.1.6 The proposed green initiatives are summarised in **Table 1.3** under the following themes: Green Mobility, Comprehensive Water Management, Solid Waste Management, Sustainable Drainage System, and Green Energy Saving.

Table 1.3 Proposed Green Initiatives

Themes	Green Initiatives
Green Mobility	Create a compact and walkable urban development by concentrating population, key economic activities and major community facilities within walking distance of public transport nodes.
	Create local communities with easily accessible daily necessities to promote walking.
	Provide comprehensive, convenient and attractive cycle track and pedestrian walkway networks with supporting facilities such as bicycle parking areas and footbridges, crossing facilities and rest areas, etc. to promote walkability and cycle friendliness. The total length of new cycle tracks within the PDA is approximately 11 km.

Themes	Green Initiatives
	Provide Employment Belt to increase the local employment opportunities and hence reduce the need for commuting traffic and subsequently the generation of vehicular noise and air pollutant emissions.
Comprehensive Water Management	Incorporate dual process of tertiary and secondary plus treatment at the proposed YLS STW allowing treated sewage effluent (TSE) to be reused as reclaimed water as much as practicable.
	Reuse of TSE as reclaimed water for non-potable uses such as toilet flushing.
Solid Waste Management	Reductions in the amount of municipal solid waste generated and disposed of through an onsite waste management plan including local recycling of organic waste, local recycling of glass for building blocks production and source separation of recyclables.
	Explore the feasibility to adopt automatic refuse collection system to reduce fuel use, odour and noise of waste transport.
Sustainable Drainage System	Provide flood retention facilities as a regulating measure to overall drainage system and as a microclimate cooling mechanism through integrated design with the open space system for public enjoyment.
	Combine flood retention facility with Hillside River Corridor with Scenic Cycle Track, which formed part of the blue-green infrastructure of the area. The Scenic Cycle Track would complement the blue-green network to maximize opportunity for public enjoyment of the natural landscape scenery and for recreation activities.
	Collect rainwater for non-potable purposes for buildings.
	Revitalise northern section of Yuen Long Nullah within the PDA by partial decking and adopt comprehensive regeneration design along the whole drainage system. The ecological system and aesthetic value along southern section of Yuen Long Nullah, Yuen Long (West) Nullah and Tin Tsuen Channel will be enhanced with pedestrian walkways introduced. The total length of nullahs revitalised is approximately 4.5 km.
	Regenerate sections of nullahs as the major green spines, breezeways and view corridors and enable better integration between different neighbourhoods.
Green Energy Saving	Encourage environmentally friendly building design and materials, and energy-saving installations such as solar hot water systems.

Themes	Green Initiatives
	Appropriate active and passive measure for energy efficient building design during construction and operation phases.
	Promote certification under BEAM Plus or other equivalent accreditations for all new buildings.
	Explore development of community gardens in open space and amenity areas to promote green living.

Benefits of the Project

1.7.1.7 The PDA contains many brownfield sites (including open storage yards, warehouses, industrial workshops, etc.) as well as villages, residential settlements, agricultural land and vacant land. These brownfield sites have created considerable environmental, traffic, visual, and other nuisance to the neighbouring communities. One of the overarching objectives of the Project is to transform these brownfield sites to more optimal uses and better land utilisation for development and promote urban-rural symbiosis. The development of the Project could result in the following direct benefits:

- **Alleviate scarcity of housing supply –** The Project will provide about 28,500 new flats, with a ratio of almost 6:4 between public housing (including public rental housing and subsidised sale flat) and private housing. This will provide a variety of housing types, responding to the needs of the community and improving the quality of life.
- **Create local job opportunities –** Approximately 10,500 new job places would be generated upon full development of the Project, through a mix of proposed industrial workshop and storage, commercial, community, and government land uses. To take advantage of the close connection with highway network and reduce environmental impacts on the residential neighbourhood, an Employment Belt is designated along YLH for storage and workshop uses. These planned uses will provide local employment opportunities as well as ease congestion at the commuting corridors between the NWNT and the urban areas. Both the existing and planned population will be benefited from these new job opportunities.
- **Commercial Provision to Meet Local Needs –** Commercial uses would be allowed on the lower floors within the residential zones, and in mixed commercial and residential developments. The mixed-use developments incorporated with PTI are planned at three locations to serve different parts of the PDA, i.e. one in each of the three Areas. These PTIs are all located near the existing villages. The commercial area would provide space for shops including street-front shops and restaurants. The commercial facilities in YLS could add vibrancy to the area and alleviate the commuting needs of YLS residents and the existing residents and villagers in the vicinity of YLS.
- **Provide spaces for Storage and Workshop –** *The Employment Belt* in Area 1 provides a major local employment cluster. About 11 ha of land are reserved for storage, workshop and open storage uses. This provides a designated area for the consolidation of brownfield activities within the PDA.
- **Provide G/IC facilities –** The Project has proposed various G/IC facilities including clinic, community hall, police station, divisional fire station and

ambulance depot, post office, sports centre and market, educational facilities, social welfare facilities, recreation facilities, etc. All these facilities will serve both existing rural communities and the planned population.

Environmental Initiatives

1.7.1.8 Environmental considerations have been the key factors in the planning of the YLS PDA. Throughout the whole formulation process of the RODP, environmental impacts as well as benefits are both identified and critically considered. Through appropriate integration of proposed land uses into the plan, the Project offers potential environmental initiatives both to conserve existing environmental resources and, where opportunities exist, to enhance and upgrade the environment on various fronts. The environmental initiatives that the Project offers are summarised below:

- Minimize industrial / residential interface;
- Conservation of egretry and other ecological resources;
- Reducing commuting traffic by providing local employment opportunities;
- Revitalisation of nullahs;
- Rezoning of degraded brownfield sites for development;
- Providing sewerage system within the PDA and STW;
- Reuse of reclaimed water;
- Providing sustainable drainage system within the PDA;
- Protection of watercourses of higher ecological value;
- Providing greening for the PDA;
- Preservation of agricultural land, secondary woodland and integration with natural greenery;
- Encouraging green commuting within and beyond the PDA;
- Clearance of existing odour sources; and
- Catering for the future e-buses and electric vehicles.

1.7.2 Minimize Industrial / Residential Interface

1.7.2.1 One of the objectives of developing the Project is to convert those brownfield sites within the PDA to more optimal uses and improve the overall environment of the area. The proposed multi-storey buildings in the Employment Belt near YLH would accommodate the brownfield operations. The Employment Belt next to the YLH will provide convenient access to the strategic highway network. Through consolidation of the brownfield operations in the Employment Belt, this will help to alleviate existing industrial/residential interface issues resulting from existing brownfield operations. Also, the freight traffic would not need to travel through the residential neighbourhood in the PDA. Also, these multi-storey buildings would serve as a physical barrier to reduce industrial noise from the existing industrial zone outside the PDA near Fui Sha Wai and the road traffic noise from YLH.

1.7.3 Conservation of Egretry and other Ecological Resources

1.7.3.1 An active egretry near Pak Sha Tsuen, woodland patches and a large piece of agricultural land near TYST, and few sections of watercourses of higher ecological

value were found within and near the PDA, and the RODP has proposed to conserve them.

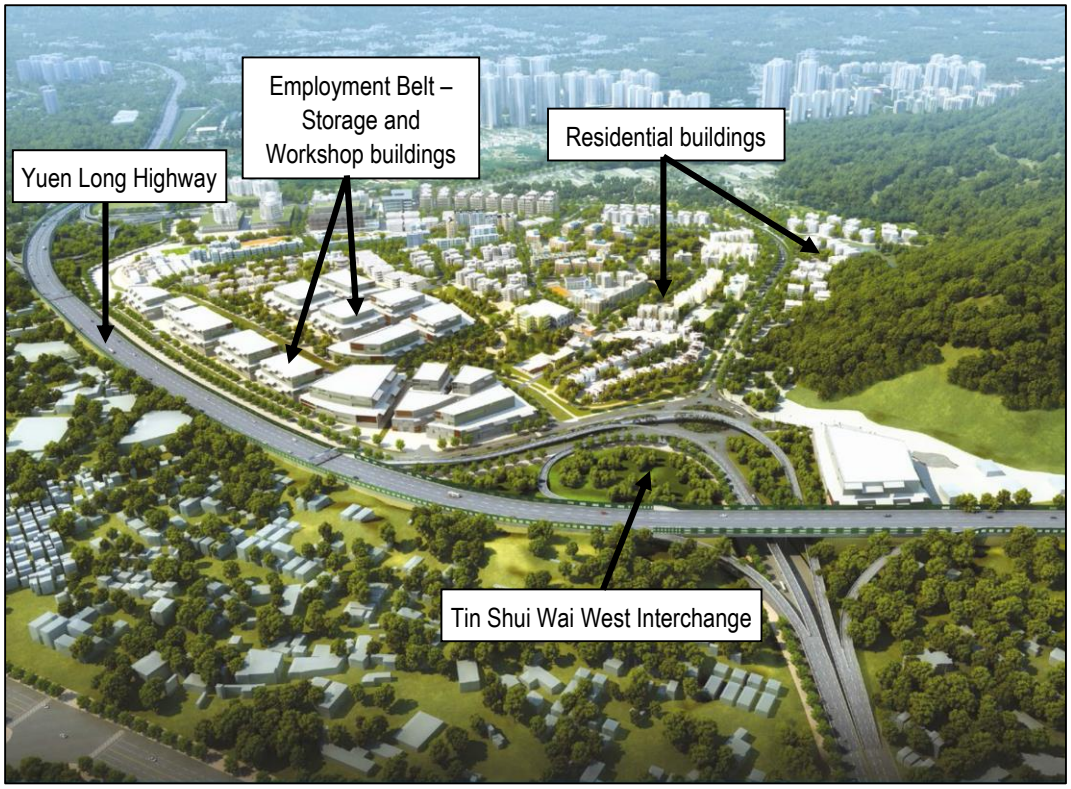
1.7.4 Reducing Commuting Traffic by Providing Local Employment Opportunities

1.7.4.1 Currently, a large area of YLS PDA of approximately 100 ha is occupied by brownfield operations such as warehouses, open storage yards, vehicle repairs, waste recycling yards and other rural industrial workshops. While it is recognised that the industrial uses provide backup services to the logistics industry and employment opportunities for locals, the intermingled land uses have been exerting various issues and nuisance on the environment, such as the noise nuisances from truck movements, odour and visual nuisance from operational activities, etc.

1.7.4.2 Taking advantage of the close connection with the highway network, the Employment Belt has been proactively proposed along YLH where the area is easily accessible to the highway and freight traffic would not need to travel through the residential neighbourhood in the PDA. The RODP has proposed zonings for the northern part of Area 1 as "Storage Use", "Storage and Workshop", "Open Storage" and "Mixed Use". Approximately 1.4 ha of land for open storage use and approximately 10 ha of land for storage and workshop uses are proposed, which are intended to support industrial and logistic operations.

1.7.4.3 Multi-storey building sites are planned at the "Storage Use" and "Storage and Workshops" zones to consolidate the existing operations on brownfield. To further segregate industrial land uses and residential areas, a network of open spaces has also been planned while the setback distance has been allowed to address road traffic noise issues, the multi-storey buildings could also serve as a screen to further reduce the impacts of road traffic noise from YLH on the residential areas in the southern part of Area 1.

1.7.4.4 It is anticipated that about 10,500 jobs will be created and these local job opportunities can help reducing need for community traffic traffic from both existing and planned population and hence the associate air emissions and traffic noise nuisance for the existing sensitive receivers along the commuting routes would be reduced.



Conceptual diagram of Employment Belt

1.7.5 Revitalisation of Nullahs

1.7.5.1 Yuen Long Nullah is a key element in the YLS, receiving water flow from hillside adjacent to Tai Lam Country Park in the south, and connecting to water channels in the town centre of Yuen Long in the north. The nullah is currently channelised and lined with concrete (approximately 2.5km long), with narrow traffic roads on both sides (i.e. Kung Um Road and Kiu Hing Road) and limited area for pedestrian passage/amenity.

1.7.5.2 As part of this Project, rehabilitation works without any decking are proposed for about 0.7 km of Yuen Long Nullah at its southern section. Along this section of the revitalised nullah, pedestrian walkway and cycle track, each with approximately 4m-wide, accompanied by amenity planting and landscaped open space have been proposed. The vegetation will introduce a green spine and improve the visual appeal of the watercourse. The revitalised nullah with soft and hard landscaping will be promoted as a place for leisure for both existing and future residents. The improvements will provide a major visual corridor for the community and help improve the micro-climate and contribute to alleviate certain urban heat island effect.

1.7.5.3 Partial decking with nullah revitalisation would be used to provide space for road safety and better pedestrian environment along the northern section of Yuen Long Nullah. Two other nullahs, i.e. Tin Tsuen Channel and Yuen Long (West) Nullah will also be revitalised. The revitalised nullahs would also help increase the food resources for the egrets.



Conceptual diagram of revitalised river channel (Overview)



Conceptual diagram of revitalised river channel

1.7.6 Rezoning of Degraded Brownfield Sites for Development

1.7.6.1 Currently, approximately 100 ha of land within the PDA is degraded brownfield sites used for industrial purpose and the original agricultural lands have already been

disturbed by brownfield operations. The brownfield area is mainly zoned as "Undetermined" and the remaining ones are zoned as "I(D)", R(D) etc. on the approved Outline Zoning Plan TYST OZP No. S/YL-TYST/10 and these areas will be zoned for other uses including open space, residential, G/IC, etc.

1.7.6.2 However, the original agricultural lands have already been disturbed by brownfield operations. Various ecological habitats have been identified within the PDA and its vicinity (see **Chapter 8** for details). They include an active egretry in Pak Sha Tsuen, woodland patches and a large piece of agricultural land near TYST, and few sections of streams within the PDA. The RODP has considered the value of these important environment receivers and has proposed to conserve them.

1.7.6.3 During the earlier planning stage, the boundaries of the PDA are designed to avoid encroachment onto these ecologically important resources as far as practicable, such as the Ecologically Important Stream (EIS) near Yeung Ka Tsuen, which is now located outside the PDA boundary. Moreover, the planned roads and road junctions have avoided the woodlands and the agricultural land as far as practicable, to avoid and minimize disturbance and fragmentation of those habitats.

1.7.6.4 Therefore, the Study has proposed rezoning of these areas, so that the ecologically important resources within these "Undetermined" land uses can be conserved. Nevertheless, there would still be approximately 5.5ha of woodland of various ecological quality that would be inevitably affected, and hence would require compensation (see **Section 8** for more details).

1.7.6.5 On this basis, landuse of the areas within the PDA near these ecologically important resources have been proactively designed to minimize human disturbance and provide additional setback where practicable. For instance, zonings of "District Open Space" ("DO"), "Local Open Space" ("LO") and "Amenity Areas" ("A") are proposed at the southern tip of Area 3 which is near Tai Lam Country Park and Yeung Ka Tsuen EIS, while "District Open Space" ("DO") and "Local Open Space" ("LO") are proposed in Area 2 near the egretry at Pak Sha Tsuen. These land uses are characterized by less human disturbance, and offer additional setback to the ecological habitats.



Aerial photo of existing brownfield sites

1.7.7 Providing Sewer System within the PDA and STW

1.7.7.1 There has been long-term improvement in Water Quality Objectives (WQO) compliance rate and Water Quality Index (WQI) grades of Hong Kong's inland water

courses (see **Section 6** for water quality baseline conditions). Despite the trend of general improvement, according to the EPD River Water Quality in Hong Kong 2014, the water quality of the water courses including those in YLS forming Yuen Long Creek have a range of Water Quality Index grading from "Bad" to "Fair".

1.7.7.2 Majority part of the PDA is currently without any centralised sewage treatment facilities, and some of the watercourses may be subject to discharges from various sources such as livestock farms, unsewered village houses and industrial establishments. The estimated existing unsewered population within the scattered villages and brownfield sites within the PDA is approximately 280 persons (about 100 households), and the estimated employment is about 4,500 persons within the existing brownfield sites within the PDA. With the proposed YLS STW implemented under the Project, all the areas within the PDA boundary will be provided with sewers that convey the sewage to the new STW. Hence, all existing unsewered population within the PDA will be sewered and the watercourses in the vicinity are anticipated to have potential benefits from the improvement in water quality for the benefit of the wider community.



Existing villages

1.7.8 Reuse of Reclaimed Water

1.7.8.1 Due to the scarce pollution credit in the Deep Bay Water Control Zone (WCZ), reuse of TSE as reclaimed water will be implemented for the PDA and adjacent developments for non-potable uses such as toilet flushing, while the surplus TSE will be discharged from YLS STW to the existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall.

1.7.8.2 A small amount of TSE will be separately discharged to the reedbed adjoining to the proposed YLS STW for further polishing before entering the revitalised Yuen Long Nullah for creation of water bodies. It has been proven that reedbed is capable to further polish the reclaimed water by means of the biological breakdown process. According to the Sewerage Impact Assessment, the reedbed removal efficiency are approximately 50% for Nitrogen, 90% for Suspended Solids (SS), 30% for 5-day Biological Oxygen Demand (BOD5) while the remaining loading could be partly compensated by the pollution credit gained from the replacement of existing livestock farms. The tertiary TSE after treatment in STW would only pass through the reedbed but not other receiving bodies such as the Hillside River Corridor, retention lake etc., before entering the revitalised Yuen Long Nullah. Hence, the TSE to be discharged to Yuen Long Nullah would not lead to a net increase in pollution loading in the Deep Bay WCZ.

1.7.9 Providing Sustainable Drainage System within the PDA

1.7.9.1 Flood retention facilities such as flood retention tank/ lake/ pond are proposed to cater for the additional runoff caused by the PDA. These storage devices retard residence time and alleviate peak flows. Instead of expelling flood water, these measures hold up water and therefore can maintain steady river flows during drier periods. These measures will also serve for amenity purposes and enhance the landscape and visual environment.

Retention tank

1.7.9.2 According to the Drainage Impact Assessment (DIA) under the Study in order to mitigate the increased flow from the development, a retention tank at Area 1 has been proposed to attenuate the peak flow, such that the peak outflow to the existing Yuen Long (West) Nullah would attain insignificant change in flow rate. The area and storage volume of the retention tank in Area 1 are about 2,000 m^2 and 10,500 m^3 respectively.

Hillside River Corridor and Retention Lake

1.7.9.3 A retention lake and Hillside River Corridor have also been proposed at the upstream of Tin Tsuen channel (along Lam Tai East/West Road) to temporarily store the 50 year peak runoff from the existing hillside catchment and catchments from proposed development in Areas 2 and 3, and hence to mitigate the increase in peak surface runoff from the PDA. The retention lake and Hillside River Corridor will also serve for amenity purposes and enhance the landscape and visual environment. According to the DIA under the Study, the area and storage volume of the retention lake are approximately 0.6ha and 13,000 m^3 respectively. The Hillside River Corridor runs along the PDA boundary and is about 575m and 0.37ha, with a storage volume of about 3,200m^3. By intercepting several hillside streams, this new watercourse will provide hydrological and ecological connectivity to downstream sections. The new watercourse will be designed to maximize ecological opportunities for aquatic flora and fauna and provide new linkages to several fragmented/isolated watercourses. Consideration should be given to the creation of an island within the Retention Lake, planted with a mixture of native trees and bamboo suitable for nesting ardeids. With appropriate planting and careful design at the detailed design stage, the Retention Lake can provide potential opportunities for waterbirds.

Reedbed/ Retention Pond

1.7.9.4 A reedbed/ retention pond of approximately 4 ha has been proposed at the south of Area 3. The reedbed can act as a sponge that traps and slowly releases surface water. Reeds can slow the speed of stormwater and distribute slowly over the area. In addition to the benefits of flood control, the permeability of reedbed allows infiltration of rain water that replenishes groundwater. This reedbed also serves as a visual element for the local community. The reedbed can provide ecological

enhancement to the area providing foraging opportunities and shelter for a suite of aquatic fauna, and in particular, wetland dependent birds.





Sustainable drainage system



Photomontage of the Hillside River Corridor and the Retention Lake

1.7.10 Protection of Watercourses of Higher Ecological Value

1.7.10.1 Three watercourses with endemic crab species (e.g. *S. zanklon*) have been identified within the PDA (see **Chapter 8** for more details). However, under the current statutory plans, these watercourses presently lack appropriate protection.

1.7.10.2 A conservation zoning of ("GB(1)") covering 15m buffer on both sides of the streams is proposed along the streams identified with endemic crab species. This would cover two sections of watercourses with a total length of about 1.6km in Area 1, and one section of about 310m in Area 3 of the PDA. Area zoned as Green Belt will be primarily conserved for its existing natural environment to safeguard it from encroachment by urban type developments. Any activities that are not favourable to the environment would be prohibited in these "GB(1)" zones and would therefore conserve the environmental conditions of these watercourses. In response to the

comment received in CE3, the channelised watercourses which are the extension of those streams/watercourses with endemic crab species are also zoned "GB(1)" with an additional intention to revitalise these channelised watercourse when opportunity arises in order to achieve habitat protection. This zone protects the habitats for wildlife from unauthorized encroachment and illegal dumping, while providing flexibility to the Town Planning Board to scrutinise developments under the planning permission system.

1.7.11 Providing Greening for the PDA

1.7.11.1 Greening is also another key consideration in formulating the RODP. According to the current proposal, a Green Zone of about 15 ha to preserve the existing agricultural land has been designated to enhance the local character, green landscape and linkage between urban and rural areas. In addition to the Green Zone, greening space/planting would be provided in the open space, amenity areas, along the Hillside River Corridor and reedbed/ retention pond.

1.7.12 Preservation of Agricultural Land, Secondary Woodland and Integration with Natural Greenery

Preservation of Green Belt and Agricultural Land

1.7.12.1 YLS was thrived with agricultural activities in the 1960's, attributed to the open and fertile plain with easily accessible water for irrigation. Despite most of the agricultural land had been converted into other land uses or become abandoned over the decades, a cluster of mixed dry and wet agricultural land (about 11 ha in total) is being retained at the southeast portion of Area 1 (the Green Zone). Besides, two patches of secondary woodland in Area 1 and the strips of land along the southwestern boundary of Area 1 which are part of woodland of the hillslope are preserved and zoned "Green Belt" ("GB").

1.7.12.2 It is proposed to preserve some of the agricultural lands with a designated zoning as "Agriculture" ("AGR") on the RODP. While there are currently small pockets of inactive agricultural land scattered within the proposed AGR zone, it is anticipated that the rezoning will facilitate agricultural rehabitilization.

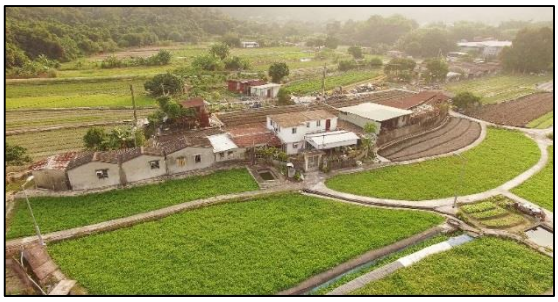
Provide Linkage between Agricultural land and Secondary Woodland

1.7.12.3 Preservation of the agricultural land would improve visual links to natural terrain and landscape features in the surround. The proposed "AGR" zone presently borders a large piece of mature secondary woodland to the south (about 18 ha), and adjoins continuous shrubland/grassland to the west which further extends to the Tai Lam Country Park. Preservation of the agricultural land and secondary woodland would improve visual links to natural terrain and landscape features in the surrounding. One of the watercourses of higher ecological value mentioned in **Section 1.7.10** also runs into the clustered agricultural land.

1.7.12.4 Efforts were made to conserve the active agricultural land and the adjacent mature secondary woodland. The active agricultural land and secondary woodland in the southern part of TYST has been identified and would be preserved by "AGR" and "GB" zonings so as to avoid incompatible use and development. Besides, the proposed road network between Areas 1 and 2 is aligned to the edge of the PDA as much as possible to minimize habitat fragmentation (see **Figure 3.1**), while avoiding intrusion to the village environs on the other side. The road alignment has been

adjusted to pass though the eastern limit of the agricultural land away from those areas with good ecological connectivity. On the other hand, the footprint of the segment of Road D1, which encroached on the agricultural land, has been reduced to minimize the disturbance to the habitat (see **Chapter 8** for more details). Through proper designation of zoning for agricultural use and conservation and due consideration to the road alignment, it is anticipated the preserved open fields with extended natural greenery would form a core Green Zone.

1.7.12.5 Existing secondary woodland and plantation would be retained as much as possible by rezoning to open space and amenity areas zonings (e.g. "A" and "DO" zones proposed next to TSWW Interchange).




Preservation of agricultural lands and provide open space and amenity areas

1.7.13 Encouraging Green Commuting and Walking within and beyond the PDA

1.7.13.1 Comprehensive networks of pedestrian walkways and cycle tracks are planned within the PDA to promote green commuting. Key pedestrian corridors will be provided with amenity planting to enhance the walking environment for pedestrians. Besides, cycle tracks of about 11km will also be built across the PDA and be connected to other districts such as Ping Shan, Tin Shui Wai New Town and Yuen Long New Town. There are two connection points between the proposed cycling network of the PDA and the external existing / planned cycle tracks. One of the connection points is along Fui Sha Wai South Road near TSWW Interchange, and the other one is along west of Long Tin Road near TYST Interchange. Cycling track and pedestrian walkway networks are also proposed to connect the PDA with Tai Lam Country Park. All these measures are proposed to facilitate walking and cycling activities across the PDA and in the wider community.

1.7.13.2 During the formulation of the RODP, space has been reserved for a possible introduction of an EFTS within the PDA. In the event that a possible EFTS being considered for the HSK NDA is extended to YLS, the extension could run along the reserved space to serve the YLS PDA. If such a system is introduced, this would give the YLS community another green transport option and more enhanced transport connection to the HSK NDA.

1.7.13.3 Currently, there is a lack of commercial and community facilities within the PDA and residents have to travel to Yuen Long New Town for their daily necessities. The RODP has planned commercial space for retail facilities and has planned G/IC facilities such as educational uses, sports and recreational facilities, elderly homes, clinic and community hall, etc. These facilities can reduce travel distance/trips of both existing and future residents of the near community to Yuen Long New Town. These provisions can also relieve the stress on public transport systems and minimize the associated environmental issues. All these facilities would be shared by both the existing rural communities and the planned populations and hence promotes the

urban-rural symbiosis which could be further enhanced through improvement of village access to the PDA.



Environmentally-friendly modes of transport

1.7.14 Clearance of Existing Odour Sources

1.7.14.1 Currently, there are six livestock farms in the YLS PDA, including three pig farms and three chicken farms. For the sake of minimising the risk of avian influenza in Hong Kong, the Government has since 2008 adopted a policy of containing the scale of live poultry trade in Hong Kong, with the total rearing capacity of live chickens capped at 1.3 million. Under the existing statutory provisions, there are constraints in relocating the affected chicken farms due to government policies.

1.7.14.2 Three livestock farms (one pig farm and two chicken farms) that are located in the central locations of the PDA have inevitably to be cleared. The chicken farms could not be relocated due to statutory restriction. In order to strike a balance among public opinions, to minimize anticipated interface issue with existing and future residents and to minimize odour impact, only one chicken farm located at the most southern fringe of the PDA is retained. Sufficient buffer area has been reserved in the RODP to mitigate for odour impact due to the retained chicken farm.

1.7.14.3 For the two pig farms located at the southern end of Kung Um Road, taking into account the adverse odour impact and the opinions of the public, the two pig farms at the southern end of Kung Um Road could not be retained under the RODP. The areas are rezoned as "G" for government depot and serve as part of the buffer between the retained chicken farm and sensitive land uses.

1.7.15 Catering for the Future E-buses and Electric Vehicles

1.7.15.1 The technological advancement in the last decade has demonstrated the feasibility of adopting more environmentally friendly buses in city environment. According to "A Clean Air Plan for Hong Kong" published by the Environmental Bureau in Year 2013 (http://www.enb.gov.hk/en/files/New_Air_Plan_en.pdf) which sets out in detail the various measures to tackle air pollution from power plants, land and sea transport, and non-road mobile machinery and to strengthen collaboration with Guangdong to deal with regional pollution. One of the options is the implementation of electric buses (e-buses), which could help to eliminate one of the major roadside sources of particulates generated by conventional buses relying on diesel fuel.

1.7.15.2 Although the feasibility and implementation issues of adopting e-buses in Hong Kong are yet to be demonstrated, it would be important to have a space provision for charging facilities within the new PTIs. According to the current landuse planning,

there will be a total of 4 PTIs – 3 within the PDA and 1 outside the PDA near Yoho Midtown. These new PTIs will be designed with sufficient space to allow for the installation of charging facilities by the bus operators in a timely manner.

1.7.15.3 Furthermore, the rapid technology development in this field has encouraged the promotion of using electric vehicles (EVs) to combat roadside air pollution problems. Apart from the assignation of charging facilities for e-buses within the new PTIs, it is also suggested to include EV charging facilities in car parks of the new private and public residential developments.

1.7.15.4 This provision in the new developments will serve as an environmental benefit to help reduce vehicular emission in the future.

1.8 Concurrent Projects

1.8.1.1 In order to assess the cumulative impacts, a review of best available information at the time of preparing this EIA report to identify a number of other projects that are undergoing planning, design, construction and/or operation within the construction and/or operation period for the Study has been conducted and a list of the tentative concurrent projects identified at this stage is summarised below and **Figure 1.3** shows the locations of these concurrent projects. Various cumulative impacts for different concurrent projects have been identified and detailed justifications on consideration of various environmental cumulative impacts from individual concurrent projects has been included in corresponding technical chapters. During the subsequent detailed design stage of YLS development, it is proposed to conduct an environmental review to collate any further available information at that time for updating the cumulative impacts, including but not limited to the implementation programme of concurrent projects such as HSK NDA.

HSK NDA Planning and Engineering Study

1.8.1.2 The development at HSK NDA will help to create new development land for the provision of approximately 61,000 new flats thereby contributing toward Hong Kong's housing supply targets. The HSK NDA project also aims to increase land supply for economic use by providing employment opportunity of about 150,000, in order to achieve the vision of building a sustainable, people-oriented and balanced living and working community for Hong Kong. This would help to address the over concentration of commercial activities and employment opportunities in the main urban areas, boost the vibrancy of local communities, meet the short fall of jobs in Tin Shui Wai New Town, as well as ease congestion at the commuting corridors between the New Territories and the urban areas.

1.8.1.3 The study commenced in mid-2011 and the final EIA Report was approved in December 2016. The project is targeted to commence the construction in 2019 with the first population intake in 2024. As the nearest works boundary of that study is located at about 450m away from the YLS PDA, air quality, noise, sewerage and ecological impact during both construction and operational phase are anticipated.

Engineering Works at Lin Cheung Road Site, Sham Shui Po and Wang Chau, Yuen Long

1.8.1.4 To support the proposed public housing development, CEDD commissioned the study for the provision of essential infrastructure. The first phase of building works of the housing flats at Wang Chau is targeted to commence in 2017/2018 and complete by 2024/2025. Another study by the Hong Kong Housing Authority (HA)

in 2012 studied the potential housing development at Wang Chau and the boundary of the project is derived from the public housing site at Wang Chau as proposed by HA.

1.8.1.5 As the works boundary of the proposed Wang Chau site is located more than 1km away from the YLS PDA, cumulative impact is not anticipated during construction phase. However, impact from vehicular emission during both construction and operational phase, and road traffic noise impact during operational phase are anticipated from the induced traffic.

Site Formation and Infrastructural Works for the Development at Long Bin, Yuen Long, Feasibility Study

1.8.1.6 The project is to conduct an engineering feasibility study to examine the feasibility of developing public housing at the potential site at Long Bin, Yuen Long; to determine the scope of site formation and infrastructure works for supporting the development; and to provide essential engineering information to facilitate the Zoning Amendment for the development site. A District Council (DC) Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10" has been discussed at the DC meeting on 21 Feb 2017.
(http://www.districtcouncils.gov.hk/yl/english/meetings/dcmeetings/dc_meetings_doc.php?year=2017&meeting_id=12080)

1.8.1.7 As mentioned in the DC Discussion Paper, the project area would be approximate 10ha for developing 11 blocks of public housing for accommodation of a population of 31,200. The construction period and population intake under this project would be divided into 2 stages, in which the population intake years of stages 1 and 2 are anticipated to be Year 2024/25 and Year 2027/28 respectively.

1.8.1.8 The project area of the development at Long Bin is located approximately 150m away from the YLS PDA opposite to YLH. Impact from vehicular emission during both construction and operational phase, and road traffic noise impact during operational phase are anticipated from the induced traffic. Cumulative impact on its planned receivers have also been assessed under this project.

Site Formation and Infrastructural Works for the Development near Tan Kwai Tsuen, Yuen Long – Feasibility Study

1.8.1.9 The scope of the Feasibility Study requires 1) to conduct necessary technical assessments for examining the feasibility of developing public housing at the sites, and for assessing preliminarily the various impacts on the existing infrastructure and environment due to the proposed development and infrastructural works; 2) to determine the scope of the supporting site formation and infrastructural works; and 3) to provide essential engineering information to support the zoning amendment application. A DC Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10" has been discussed at the DC meeting on 21 Feb 2017. (http://www.districtcouncils.gov.hk/yl/english/meetings/dcmeetings/dc_meetings_doc.php?year=2017&meeting_id=12080)

1.8.1.10 As mentioned in the DC Discussion Paper, the project area would be approximate 10.6ha for developing 7 blocks of public housing for accommodation of a population of 20,600, and the population intake year is anticipated to be Year 2027/28.

1.8.1.11 The project area of the development at Tan Kwai Tsuen is located approximately 300m away from the YLS PDA next to the Tan Kwai Tsuen Fresh Water Service Reservoir along YLH. Impact from vehicular emission during both construction and operational phase and cumulative impact on its planned receivers have also been assessed under this project. Road traffic noise impact during operational phase are anticipated from the induced traffic.

Improvement to Pok Oi Interchange

1.8.1.12 To improve the traffic capacity of Pok Oi Interchange to relieve the existing traffic pressure and meet the future traffic demand. The project includes provision of an additional lane of YLH (Kowloon bound) at Pok Oi Interchange, construction of a slip road connecting the elevated and ground level sections of YLH (Tuen Mun bound), and construction of a segregated left-turn lane at the approach road of Pok Oi Roundabout near Pok Oi Hospital.

1.8.1.13 The scheme commenced in late 2012 and is completed in 2016. Cumulative impact during construction is not anticipated. The impact from vehicular emission and road traffic noise impact during operational phase are anticipated from the induced traffic.

Elevated Pedestrian Corridor in Yuen Long Town Connecting with Long Ping Station – Investigation, Design and Construction

1.8.1.14 The purpose of the project is to construct an elevated pedestrian corridor above Yuen Long Town Nullah from West Rail Long Ping Station crossing over Yuen Long On Ning Road, Castle Peak Road - Yuen Long Section to the south of Kau Yuk Road with provision for future extension to Yuen Long South areas. The corresponding EIA report has been prepared and approved by EPD in Oct 2016 (AEIAR-200/2016).

1.8.1.15 The scheme will tentatively commence in 2018 and is targeted to be completed by 2022. Cumulative impact on air quality and water quality during construction is anticipated. Noise impact during construction phase and operational phase are not anticipated.

Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 1 Improvement Works – Design and Construction

1.8.1.16 The project comprises improvement works for the Town Centre Section of the Yuen Long Town Nullah by intercepting the polluted dry weather flow for treatment at the YLSTW and re-using the treated water to provide scenic water flow for flushing the Town Centre Section of the Yuen Long Nullah. Approximately 60 existing storm water outfalls within the town centre section will be intercepted by the proposed system.

1.8.1.17 According to the information from DSD website, the construction of the improvement works commenced in Year 2022 and is targeted to be completed in Year 2026. Cumulative impact is not anticipated during the construction phase. No impact is anticipated during the operational phase as well.

Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 2 Beautification Works – Design and Construction

1.8.1.18 The project involves beautification and landscaping works by modifying and reconstructing the existing concrete nullah bed and wall along the 800m town centre section of the Yuen Long Town Nullah.

1.8.1.19 According to the relevant government department, the construction of the development will be commenced in Year 2027 and is targeted to be completed in Year 2029. Cumulative air quality and water quality impact is anticipated during the construction phase. During the operational phase, no impact is anticipated.

Design-Build-and-Operate contract for upgrading San Wai Sewage Treatment Works – Phase 1

1.8.1.20 The project comprises the design and construction of sewage treatment facilities with a daily treatment capacity of 200,000 m^3 to cater for the projected additional sewage flow due to development needs in the Northwest New Territories after 2020. The works will also upgrade the treatment level of the STW to the chemically enhanced primary treatment level with ultraviolet disinfection facilities for reducing pollution loads to the northwestern waters.

1.8.1.21 According to the information from DSD website, the construction of the development commenced in mid Year 2016 and is targeted to be completed in late Year 2020. As the upgrading works are located over 2km away from YLS development, cumulative impact during construction phase is not anticipated. Nevertheless, cumulative effect from sewerage and sewage treatment implications is anticipated during the operation phase.

Cycle Tracks Connecting North West New Territories with North East New Territories – Tuen Mun to Sheung Shui section (Stage 1)

1.8.1.22 This project comprises construction of approximately 2.5 km new cycle tracks along Kam Tin River in Yuen Long and along Shek Sheung River and Sheung Yue River in Sheung Shui, improvement works to about 4.5 km in length of selected sections of existing cycle tracks in Tuen Mun, Tin Shui Wai and Yuen Long and construction of entry/exit hubs and resting stations along the cycle tracks.

1.8.1.23 The scheme commenced in late 2013 and has been substantively completed in December 2016. Cumulative impact during both construction and operational phases is not anticipated.

Property Development of Kam Sheung Road Station and its Adjoining Areas

1.8.1.24 A Land Use Review (LUR) has been conducted to examine the development potential of the Kam Tin South and Pat Heung Areas. The LUR has demonstrated that it would be technically feasible to transform Kam Tin South and Pat Heung into a sustainable suburban township with enhancement on infrastructure and utility services.

1.8.1.25 As the area is located more than 1km away from the YLS PDA, cumulative impact is not anticipated during the construction phase. However, impact from vehicular emission and road traffic noise impact during the operational phase are anticipated from the induced traffic.

Greening Master Plan (GMP) for Yuen Long - Greening Works in Tuen Mun and Yuen Long

1.8.1.26 The captioned study aims to formulate GMPs for Yuen Long District. Baseline studies and necessary site investigation were conducted, and detailed design of associated greening measures were proposed. The study commenced in late 2014 and was scheduled to complete in 2017.

1.8.1.27 Cumulative impact during the construction and operational phases is not anticipated.

Table 1.3 Potentially concurrent projects within local vicinity of the proposed development

Concurrent Project	Project Proponent	Construction commenced / due to commence	Operation/Completion of development	Potential Cumulative Impacts	
				Construction Phase	Operational Phase
HSK NDA Planning and Engineering Study	CEDD	2019	2024 (First year of population intake)	• Air Quality; • Noise; • Sewerage; • Ecology	• Air Quality; • Noise; • Sewerage
Engineering Works at Lin Cheung Road Site, Sham Shui Po and Wang Chau, Yuen Long	CEDD	Mid 2018	2024/25 (First year of population intake)	-	• Air Quality; • Noise
Site Formation and Infrastructural Works for the Development at Long Bin, Yuen Long, Feasibility Study	CEDD	N/A	2024/25 (First year of population intake) 2027/28 (Second year of population intake)	-	• Noise
Site Formation and Infrastructural Works for the Development near Tan Kwai Tsuen, Yuen Long – Feasibility Study	CEDD	N/A	2027/28 (Second year of population intake)	-	• Noise
Improvement to Pok Oi Interchange	CEDD	Late 2012	Early 2016	-	• Air Quality; • Noise
Elevated Pedestrian Corridor in Yuen Long Town Connecting with Long Ping Station – Investigation, Design and Construction	CEDD	Early 2018	Mid 2022	• Air Quality; • Water Quality	-
Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 1 Improvement Works – Design and Construction	DSD	2022	2026	-	• Sewerage
Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 2 Beautification Works – Design and Construction	DSD	2027	2029	• Air Quality; • Water Quality	-

Concurrent Project	Project Proponent	Construction commenced / due to commence	Operation/Completion of development	Potential Cumulative Impacts	
				Construction Phase	Operational Phase
Design-Build-and-Operate contract for upgrading San Wai Sewage Treatment Works – Phase 1	DSD	Mid 2016	Late 2020	-	• Sewerage
Cycle Tracks Connecting North West New Territories with North East New Territories – Tuen Mun to Sheung Shui section (Stage 1)	CEDD	Late 2013	End 2016	-	-
Property Development of Kam Sheung Road Station and its adjoining areas	PlanD	Phase 1 development tentatively schedule to be tendered in 2016/2017	Tentatively in 2025 (First year of population intake)	-	• Air Quality; • Noise
Greening Master Plan (GMP) for Yuen Long - Greening Works in Tuen Mun and Yuen Long	CEDD	End 2014	Mid 2017	-	-

Note:
[1] "-" Potential cumulative impact not anticipated; "N/A" – information not available.

1.9 Objectives of the EIA Study

1.9.1.1 The objectives of this EIA Study are as follows:

- To describe the Project and associated works together with the requirements and environmental benefits for carrying out the Project and associated works;
- To identify and describe elements of the community and environment likely to be affected by the Project and associated works and/or likely to cause adverse impacts to the Project, including both the natural and man-made environment and associated environmental constraints;
- To provide information on the consideration of alternatives including, but not limited to, location, layout, scale, scope, design, construction method and mode of operation (including green transportation within the PDA), with a view to avoiding and minimizing the potential environmental impacts to environmentally sensitive areas and other sensitive uses; to compare the environmental benefits and dis-benefits of different alternatives; to provide reasons for selecting the preferred option(s); and to describe the part environmental factors played in the selection of the preferred option(s);
- To identify and quantify all environmental sensitive receivers, emission sources and determine the significance of impacts on sensitive receivers and potential affected uses and to propose measures to mitigate these impacts with respect to air quality, noise, water quality, waste, land contamination and landscape and visual;
- To identify and quantify any potential losses or damage to flora, fauna and wildlife habitats and to propose measures to mitigate these impacts;
- To identify and quantify any potential fisheries impact and to propose measures to mitigate these impacts;
- To identify any negative impacts on cultural heritage and to propose measures to mitigate these impacts;
- To identify and systematically evaluate any potential landscape and visual impacts and to propose measures to mitigate these impacts;
- To propose mitigation measures to minimize pollution, environmental disturbance and nuisance during the construction and operation of the Project;
- To investigate the feasibility, practicability, effectiveness and implications of the proposed mitigation measures;
- To identify, predict and evaluate the residual environmental impacts (i.e. after practicable mitigation) and the cumulative effects expected to arise during the construction and operation of the Project in relation to the sensitive receivers and potentially affected uses;
- To identify, assess and specify methods, measures and standards, to be included in the detailed design, construction and operation of the Project which are necessary to mitigate these environmental impacts and cumulative effects and reduce them to acceptable levels;
- To investigate the extent of the secondary environmental impacts that may arise from the proposed mitigation measures and to identify constraints associated with the mitigation measures recommended in the EIA Study, as well as the provision of any necessary modification;

- To identify, within the Study Area, any individual projects(s) that fall under Schedule 2 of the EIA Ordinance; to ascertain whether the findings of this EIA Study have adequately addressed the environmental impacts of those projects; and where necessary, to identify the outstanding issues that need to be addressed in any further detailed EIA Study; and
- To design and specify environmental monitoring and audit requirements to check the effective implementation of the recommended environmental protection and pollution control measures.

1.10 Structure of EIA Report

1.10.1.1 The structure of this technical report is as follows :

Section	Title	Aims
1.	Introduction	Introduces the project background, objectives of this report. Discusses the environmental benefits of the project
2.	Consideration of Alternative Development Options	Summarizes the consideration and selection of alternative options for development
3.	Project Description and Construction Methodologies	Describes the project need, construction methodologies and detailed elements of the development
4.	Air Quality Impact	Presents the legislation, methodology, assessment and recommendations for air quality impacts
5.	Noise Impact	Presents the legislation, methodology, assessment and recommendations for noise impacts
6.	Water Quality Impact	Presents the legislation, methodology, assessment and recommendations for water quality impact
7.	Sewerage and Sewage Treatment Implications	Presents the legislation, methodology, assessment and recommendations for sewerage and sewage treatment
8.	Ecological Impact	Presents the legislation, methodology, assessment and recommendations for ecological impacts
9.	Fisheries Impact	Presents the legislation, methodology, assessment and recommendations for fisheries impacts
10.	Landscape and Visual Impact	Presents the legislation, methodology, assessment and recommendations for landscape and visual impacts
11.	Waste Management Implications	Presents the legislation, methodology, assessment and recommendations for waste management
12.	Land Contamination Impact	Presents the legislation, methodology, assessment and recommendations for land contamination evaluation

Section	Title	Aims
13.	Cultural Heritage Impact	Presents the legislation, methodology, assessment and recommendations for cultural heritage impacts
14.	Environmental Monitoring & Audit Requirements	Presents the EM&A requirements
15.	Summary of Environmental Outcome	Presents a summary of the key environmental outcomes arising from the EIA Study
16.	Conclusion	Summarizes the findings and concludes the overall acceptability of the project